RIKESH MANDAL

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PROFESSIONAL PROFILE

MSc Artificial Intelligence graduate (Distinction, 82%) with a First-Class Honours degree in Computer Networking and IT Security, and professional experience in software development within the FinTech industry. Skilled in Python, Java, Spring Boot, REST APIs, SQL, Docker, and PyTorch, with hands-on experience developing backend applications, machine learning models, and full-stack web solutions. Brings an interdisciplinary background in software engineering, AI, and 3D computer graphics, with strong problem-solving abilities and experience translating technical requirements into practical solutions. Passionate about building reliable, scalable software and applying emerging technologies to real-world challenges.

KEY SKILLS

Programming Languages: Java, Python, JavaScript, MATLAB

Backend & Web Development: Spring Boot, Flask, Django, REST APIs, Java Spark, Jinja2, HTML, CSS

APIs & Messaging: REST API Design, Swagger/OpenAPI, Postman, JSON, NATS, Event-Driven Architecture, WebSockets

Databases: MySQL, PostgreSQL, SQLite, MongoDB, SQLAlchemy, Database Indexing, Query Optimisation

Frontend: Vue.js, HTML, CSS, JavaScript

Testing & QA: Unit Testing, Integration Testing, Pytest, Test Case Design, Bug Reproduction & Documentation, Postman

CI/CD & DevOps: Git, GitHub, CI/CD Pipelines, Docker (foundational), Linux, AWS (Academy Cloud Foundations)

Machine Learning & AI: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Open3D

Databases: MySQL, PostgreSQL, SQLite, MongoDB

PROFESSIONAL EXPERIENCE

Assistant Java Developer, FocusOne Payment Solutions (Fintech) **2022 – 2024**

Backend systems powering real-time credit and debit card authorisation for a regulated fintech platform

- Designed and maintained high-performance REST APIs using Java and Spring Boot for a platform processing real-time card authorisations across thousands of daily transactions.
- Refactored legacy applications through root-cause debugging and targeted code improvements, reducing production incidents and improving system reliability.
- Evaluated NATS messaging for event-driven architecture adoption and presented findings that shaped the team's infrastructure roadmap for low-latency, fault-tolerant systems.
- Built and maintained Swagger/OpenAPI documentation and contributed to API endpoint testing using Postman, cutting onboarding time and improving cross-team collaboration on API contracts.
- Delivered new backend features aligned with evolving product requirements in a fast-paced Agile environment, participating in code reviews and sprint planning.
- Authored system specifications and API design documentation supporting long-term maintainability, reliability, and regulatory compliance.

KEY PROJECTS

Microblog — Full-Stack Flask Web Application | Github| Live Demo | API Documentation

Developed and deployed a Flask web application with REST API design, containerized infrastructure, OpenAPI documentation, full-text search, multilingual translation and cloud deployment.

- Designed and implemented REST API endpoints for users, posts, messages, followers/following and token management, using HTTP Basic Authentication for token issuance, Bearer token

authentication for protected routes, JSON request/response handling, pagination, and HTTP status codes.

- Modelled relational data with SQLAlchemy ORM and MySQL, implementing foreign-key relationships, migrations with Flask-Migrate/Alembic, database indexing, query optimisation and reusable paginated API responses.
- Containerised the application with Docker and Docker Compose, orchestrating Flask/Gunicorn, MySQL, Elasticsearch and LibreTranslate services with internal service discovery, environment-based configuration and persistent named volumes for database, search-index and translation model data.
- Integrated Elasticsearch for full-text search, including authenticated HTTPS communication, mounted TLS certificates, CA verification and reindexing of application data when search indexes are recreated.
- Integrated LibreTranslate and Flask-Babel for multilingual post translation and internationalisation, with LibreTranslate running as an independently containerised service within the Compose network.
- Authored and validated an OpenAPI 3.0 specification covering authentication, request bodies, pagination, reusable schemas, error responses and API contracts. Rendered the documentation with Scalar and published it through GitHub Pages.
- Deployed the Flask application and MySQL database to Railway using production environment variables and Gunicorn; retained Elasticsearch and LibreTranslate as local/self-hosted services due their higher storage and memory requirements.

Distributed Messaging System with NATS.io (Java Microservices)

- Led the design and implementation of a distributed microservices communication system using NATS.io, taking responsibility for architecture, message design, and system integration strategy.
- Built pub-sub based request-reply workflows in Java, handling concurrent request processing via multi-threaded responders to improve responsiveness under load.
- Implemented queue groups to balance requests across multiple service instances, improving reliability and preventing bottlenecks.
- Defined JSON-based message contracts enabling independently built services to communicate without tight coupling, and tested the system under concurrent load to validate stability.

Monocular Image-to-3D Reconstruction (MSc Dissertation) **Github**

- Built an end-to-end deep learning pipeline using a ResNet-50 encoder-decoder for monocular depth estimation, converting depth maps to 3D point clouds via Open3D with Poisson surface reconstruction and mesh refinement.
- Tech stack: Python, PyTorch, OpenCV, Open3D, NumPy.

Sign Language-to-Speech Glove **Github**

- Built a hardware-software IoT system using an Arduino glove with flex sensors and a gyroscope to capture hand gestures, and trained a KNN classifier on a self-collected dataset to translate ASL gestures into real-time speech.
- Tech stack: Python, Arduino, Scikit-learn, IoT sensors.

Animal Image Classification System **Github**

- Designed a custom CNN from scratch (no pre-trained weights) to classify 9 animal classes, achieving 79.51% accuracy using progressive convolutional depth, 3x3 filters, batch normalisation, and dropout.
- Tech stack: Python, PyTorch, OpenCV, NumPy, Scikit-learn.

EDUCATION

MSc Artificial Intelligence — Distinction (82%) **2024 – 2025**
University of East London

BSc Computer Networking & IT Security — First Class (81.94%) **2021 – 2023**
London Metropolitan University

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- Google Data Analytics Specialisation — Coursera (2023)
- API Designer & API Security Architect — API Academy
- AWS Academy Cloud Foundations — Amazon Web Services
- Deep Learning Onramp & Machine Learning Onramp — MathWorks
- Data Science Foundations — LinkedIn Learning